

*Heaven's Light is Our Guide*



**Rajshahi University of Engineering and Technology**  
**Dept. of Electrical and Computer Engineering**

**Course Title:** Digital Techniques Sessional  
**Course Code:** ECE-2112

**Lab Report 02**

**Report Title:** Simplification of Boolean Expressions Using Boolean Algebra  
and Karnaugh Map, and Verification Using Logic Circuit Simulation

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| Submitted To:                                                                                    | Submitted By:                                                           |
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## Problem: 01

**Problem Definition:** Simplification of the Boolean expression  $F(A,B,C) = A'BC + AB'C + ABC' + ABC$  using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

### Objective:

To simplify a three-variable Boolean function from its sum-of-minterms representation into a minimum sum-of-products form, validate the simplification using a Karnaugh map, and compare the original and reduced logic circuits in Logisim to ensure they produce identical outputs.

### Algebraic Simplification:

$$\begin{aligned} F(A,B,C) &= A'BC + AB'C + ABC' + ABC \\ &= A'BC + ABC + AB'C + ABC + ABC' + ABC \\ &= BC(A' + A) + AC(B' + B) + AB(C' + C) \\ &= BC + AC + AB \\ &= AB + AC + BC \end{aligned}$$

### Truth Table:

| A | B | C | $F = AB + AC + BC$ |
|---|---|---|--------------------|
| 0 | 0 | 0 | 0                  |
| 0 | 0 | 1 | 0                  |
| 0 | 1 | 0 | 0                  |
| 0 | 1 | 1 | 1                  |
| 1 | 0 | 0 | 0                  |
| 1 | 0 | 1 | 1                  |
| 1 | 1 | 0 | 1                  |
| 1 | 1 | 1 | 1                  |

Table 1: Truth table for  $F = AB + AC + BC$

### Karnaugh Map:

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 0  | 0  | 1  | 0  |
| 1                 | 0  | 1  | 1  | 1  |

Table 2: K-map for  $F(A,B,C) = A'BC + AB'C + ABC' + ABC$

Three pairs of adjacent 1-cells are grouped in the Karnaugh map. The first pair corresponds to  $BC = 11$ , giving the term  $BC$ . The second pair covers the cells where  $A = 1$  and  $C = 1$ , resulting in  $AC$ . The third pair contains the cells where  $A = 1$  and  $B = 1$ , giving  $AB$ . Therefore, the simplified Boolean expression is  $F = AB + AC + BC$ , which agrees with the result obtained through algebraic simplification.

### Circuit Diagram:

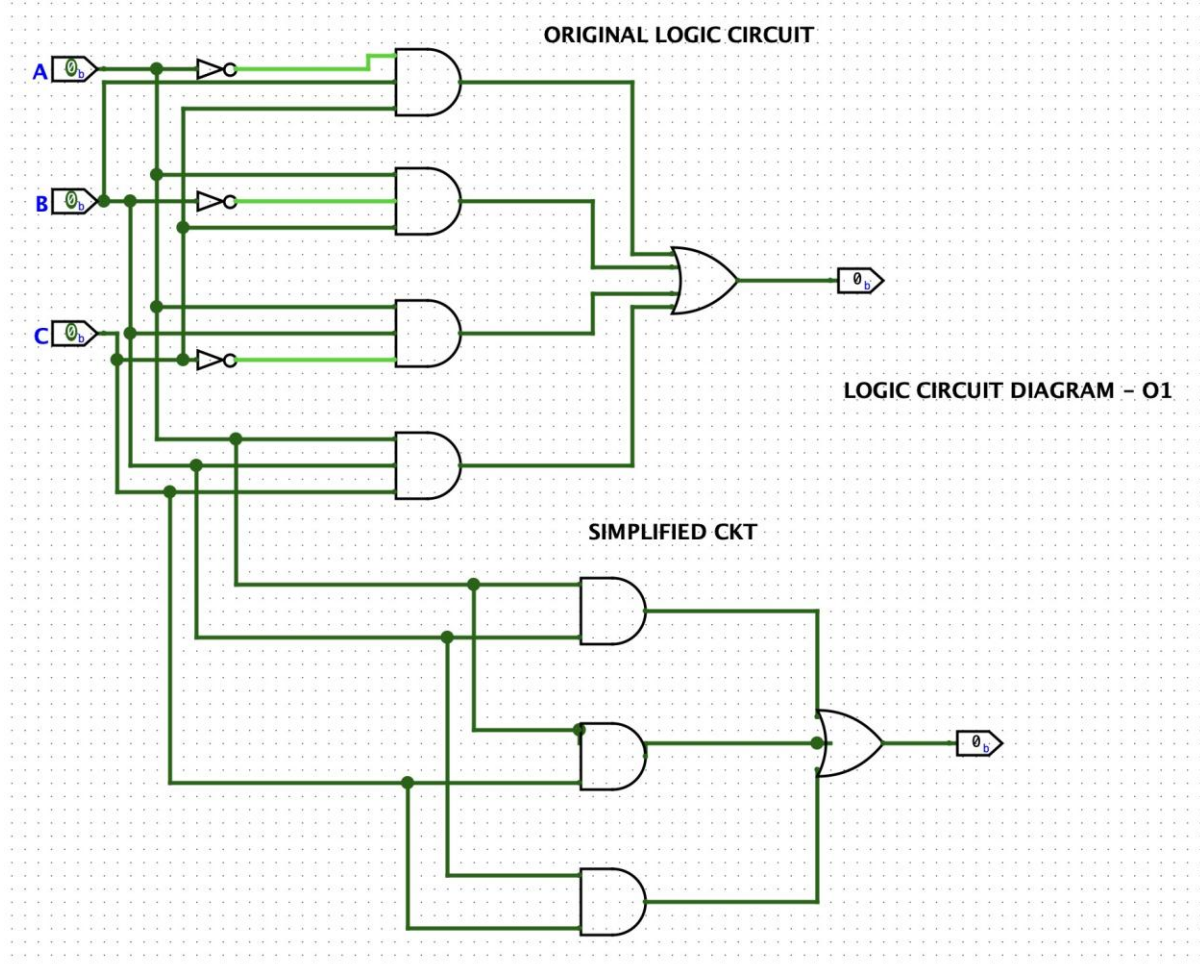


Figure 1: Original circuit (four 3-input AND gates plus a 3-input OR gate, realizing  $A'BC + AB'C + ABC' + ABC$ ) and simplified circuit (three 2-input AND gates plus an OR gate, realizing  $AB + AC + BC$ ), both simulated in Logisim

### Discussion:

The original circuit uses four 3-input AND gates, inverters, and a 3-input OR gate. The simplified circuit uses three 2-input AND gates and one OR gate. Both circuits were tested in Logisim and produced the same outputs for all input combinations, confirming the simplification. The simplified circuit is more efficient, requiring fewer gates and no inverters.

**Conclusion:**

The expression  $F = A'BC + AB'C + ABC' + ABC$  was simplified to  $F = AB + AC + BC$  using Boolean algebra and verified with a Karnaugh map. Logisim simulations of both circuits produced identical outputs for all input combinations, confirming the simplification.

**Problem: 02**

**Problem Definition:** Simplification of the Boolean expression  $F(A,B,C) = A(A + B)(A+B +C)$  using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

**Objective:**

To simplify a three-variable product-of-sums Boolean expression, verify the simplified form using a Karnaugh map, and confirm the equivalence of the original and simplified circuits through Logisim simulation.

**Algebraic Simplification:**

$$\begin{aligned}
 F(A,B,C) &= A(A + B)(A + B + C) = (A + AB)(A + B + C) \\
 &= A(A + B + C) + AB(A + B + C) \\
 &= A + AB + AC + AB + AB + ABC \\
 &= A + AB + AC + ABC \\
 &= A(1 + B + C + BC) \\
 &= A
 \end{aligned}$$

**Truth Table:**

| $A$ | $B$ | $C$ | $F = A$ |
|-----|-----|-----|---------|
| 0   | 0   | 0   | 0       |
| 0   | 0   | 1   | 0       |
| 0   | 1   | 0   | 0       |
| 0   | 1   | 1   | 0       |
| 1   | 0   | 0   | 1       |
| 1   | 0   | 1   | 1       |
| 1   | 1   | 0   | 1       |
| 1   | 1   | 1   | 1       |

Table 3: Truth table for  $F = A$

**Karnaugh Map:** All four cells in the  $A = 1$  row are 1, while the entire  $A = 0$  row is 0. This forms a single group of four adjacent cells covering the whole row, which eliminates both  $B$  and  $C$  and leaves only  $F = A$ , exactly matching the algebraic simplification.

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 0  | 0  | 0  | 0  |
| 1                 | 1  | 1  | 1  | 1  |

Table 4: K-map for  $F(A,B,C) = A(A + B)(A + B + C)$

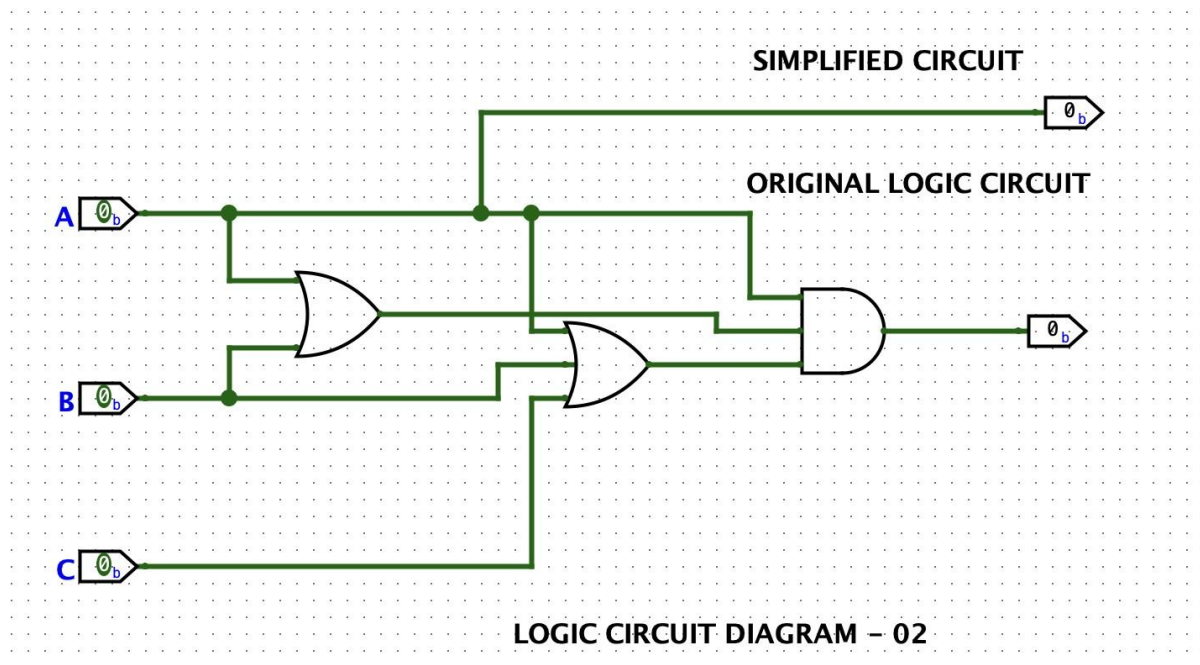


Figure 2: Original circuit (two OR gates realizing  $A+B$  and  $A+B+C$ , combined with  $A$  through an AND gate) and simplified circuit (a direct connection representing  $F = A$ ), simulated in Logisim

### Discussion:

The original circuit uses OR gates to form  $A + B$  and  $A + B + C$ , with both results combined with  $A$  through AND gates. Using the absorption law, the expression simplifies to  $A$ . Therefore, the simplified circuit requires only a direct connection from  $A$  to the output. Logisim testing confirmed that the output always matches  $A$  for all input combinations.

**Conclusion:**

The expression  $F = A(A+B)(A+B+C)$  was shown to reduce to  $F = A$  through repeated application of the absorption law, and this result was confirmed by both the Karnaugh map and Logisim simulation, which showed that  $B$  and  $C$  have no effect whatsoever on the output.

**Problem: 03**

**Problem Definition:** Simplification of the Boolean expression  $F(A,B,C) = (A + (BC)')(AB + ABC)$  using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

**Objective:**

To simplify a mixed AND-OR-complement Boolean expression using De Morgan's theorem and Boolean algebra, verify the result with a Karnaugh map, and confirm through Logisim that the simplified circuit produces a constant output.

**Algebraic Simplification:**

$$\begin{aligned}
 F(A,B,C) &= (A + (BC)')(AB + ABC) \\
 &= A' \cdot BC \cdot (AB + ABC) && \text{(De Morgan's theorem)} \\
 &= A'BC \cdot AB(1 + C) \\
 &= A'BC \cdot AB \\
 &= (AA')BCB \\
 &= 0
 \end{aligned}$$

**Truth Table:**

| $A$ | $B$ | $C$ | $F = 0$ |
|-----|-----|-----|---------|
| 0   | 0   | 0   | 0       |
| 0   | 0   | 1   | 0       |
| 0   | 1   | 0   | 0       |
| 0   | 1   | 1   | 0       |
| 1   | 0   | 0   | 0       |
| 1   | 0   | 1   | 0       |
| 1   | 1   | 0   | 0       |
| 1   | 1   | 1   | 0       |

Table 5: Truth table for  $F = 0$

### Karnaugh Map:

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 0  | 0  | 0  | 0  |
| 1                 | 0  | 0  | 0  | 0  |

Table 6: K-map for  $F(A,B,C) = (A + (BC)')'(AB + ABC)$

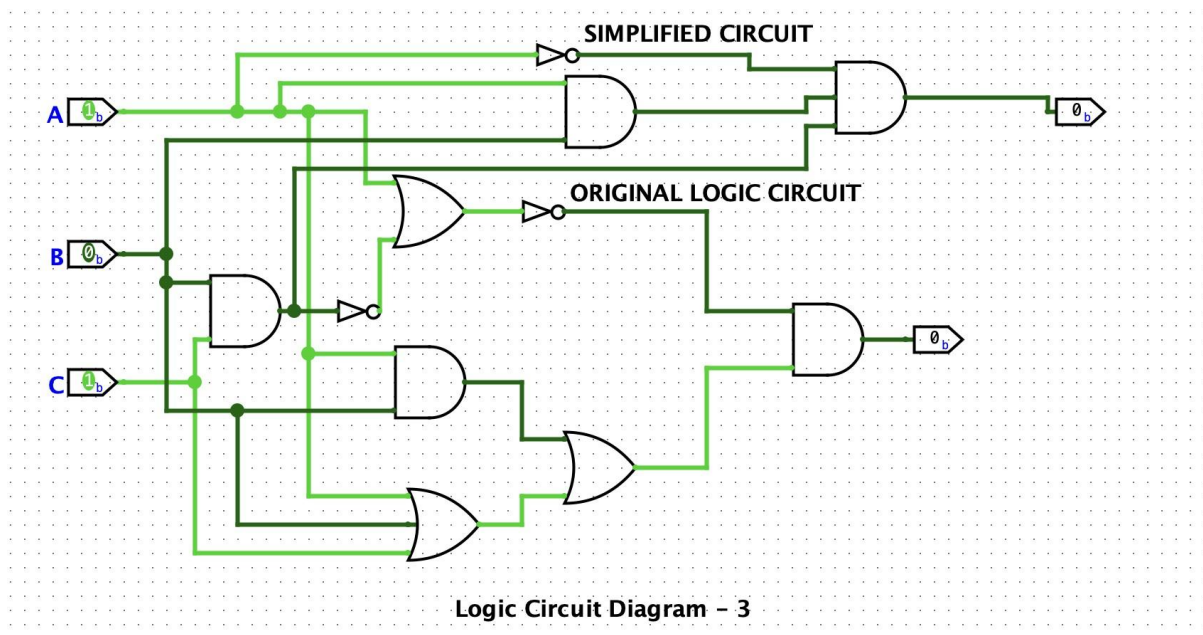


Figure 3: Original circuit realizing  $(A + (BC)')'(AB + ABC)$  using NAND-type and OR/AND stages, and simplified circuit showing the output permanently at logic 0, simulated in Logisim

### Discussion:

Using De Morgan's theorem, the first part simplifies to  $A'BC$ . The second factor  $AB + ABC$  reduces to  $AB$  by absorption. Since the result contains  $A'A = 0$ , the entire expression simplifies to 0. Logisim testing confirmed that the output remains 0 for all input combinations, matching the Karnaugh map.

### Conclusion:

The expression  $F = (A + (BC)')'(AB + ABC)$  was simplified to 0 using Boolean algebra. The result was confirmed by an empty Karnaugh map and Logisim simulation, which showed a constant logic-0 output for all input combinations.

## Problem: 04

**Problem Definition:** Simplification of the Boolean expression  $F(A,B) = (B'(A+B) + (A+B)(A+B'))B'$  using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

### Objective:

To reduce a two-variable expression built from nested OR and AND terms to its minimal form, to confirm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

### Algebraic Simplification:

$$\begin{aligned} F(A,B) &= (B'(A+B) + (A+B)(A+B'))B' \\ &= (AB' + BB' + AA + AB' + AB' + BB')B' \\ &= (A + AB' + AB' + AB')B' \\ &= A(1 + B' + B' + B')B' \\ &= AB' \end{aligned}$$

### Truth Table:

| A | B | $F = AB'$ |
|---|---|-----------|
| 0 | 0 | 0         |
| 0 | 1 | 0         |
| 1 | 0 | 1         |
| 1 | 1 | 0         |

Table 7: Truth table for  $F = AB'$

### Karnaugh Map:

| A \ B | 0 | 1 |
|-------|---|---|
| 0     | 0 | 0 |
| 1     | 1 | 0 |

Table 8: K-map for  $F(A,B) = (B'(A+B) + (A+B)(A+B'))B'$

Only a single cell, at  $A = 1, B = 0$ , is populated. A lone 1-cell cannot be combined with any adjacent cell, so it is read directly as the product of its coordinate literals, giving  $F = AB'$ , matching the algebraic result.



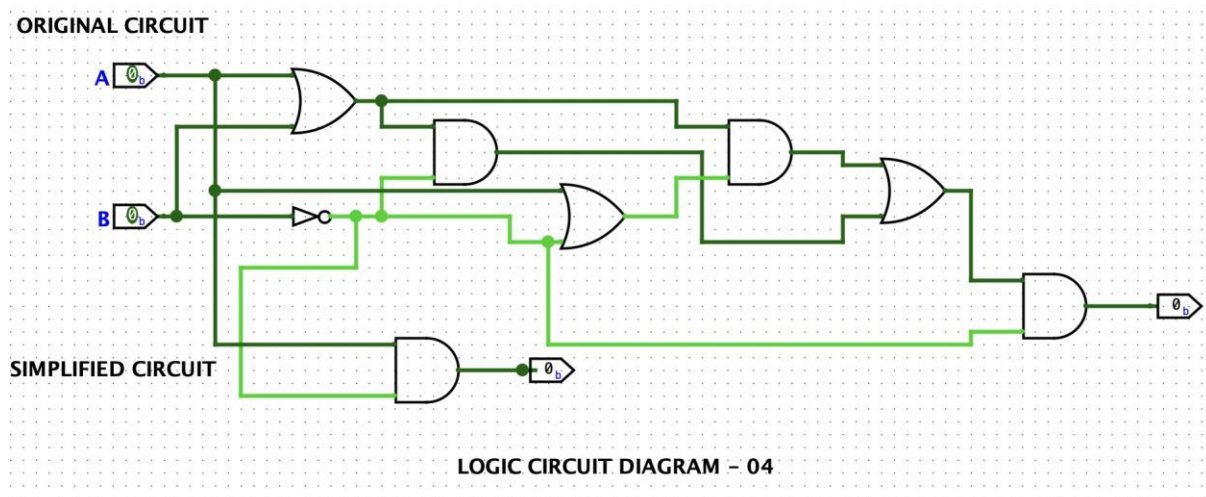


Figure 4: Original circuit realizing  $(B'(A+B) + (A+B)(A+B'))B'$  using OR, AND, and NOT stages, and simplified circuit reduced to a single AND gate realizing  $AB'$ , simulated in Logisim

### Discussion:

The original expression simplifies to  $F = AB'$  after applying Boolean algebra. Thus, the original circuit is equivalent to a simple 2-input AND gate with B inverted. Logisim testing for all input combinations confirmed that the output is 1 only when  $A = 1$  and  $B = 0$ , verifying the simplification.

### Conclusion:

The expression  $F = (B'(A+B) + (A+B)(A+B'))B'$  was successfully reduced to  $F = AB'$  using Boolean algebra, confirmed with a Karnaugh map, and validated by Logisim simulation, which showed identical outputs for the original and simplified circuits across all input combinations.

## References

1. M. M. Mano, *Digital Design*, 5th ed. Upper Saddle River, N.J.: Pearson, 2013.
2. R. J. Tocci, N. S. Widmer, and G. L. Moss, *Digital Systems: Principles and Applications*, 11th ed. Upper Saddle River, N.J.: Pearson, 2011.